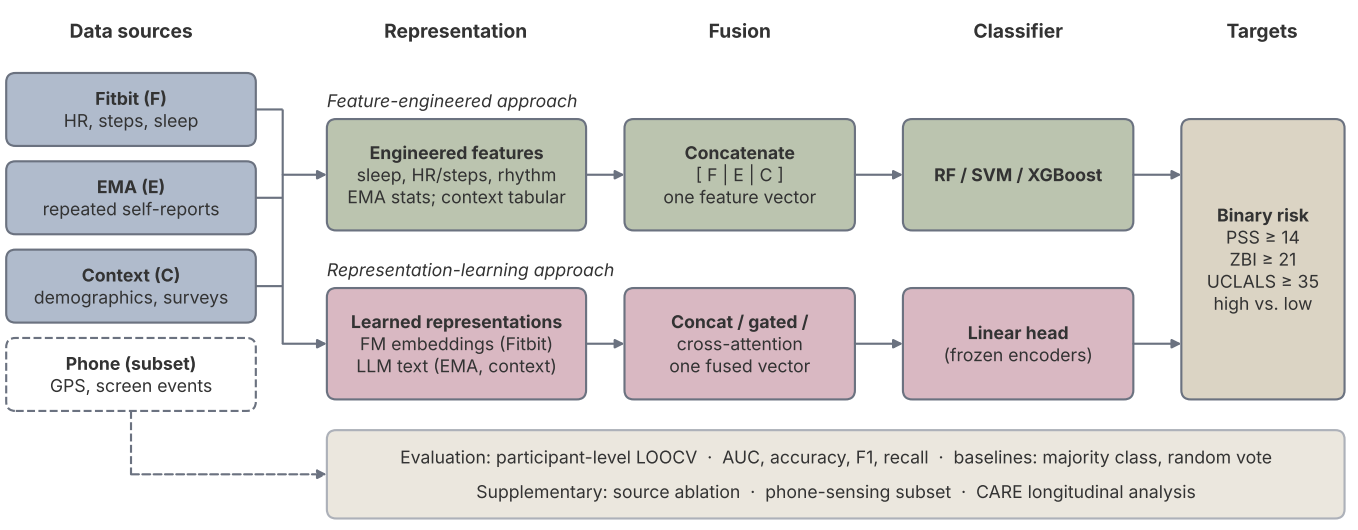
Data sources
Representation
Fusion
Classifier
Targets
Fitbit (F)
HR, steps, sleep
EMA (E)
repeated self-reports
Context (C)
demographics, surveys
Phone (subset)
GPS, screen events
Feature-engineered approach
Engineered features
sleep, HR/steps, rhythm
EMA stats; context tabular
Concatenate
[ F | E | C ]
one feature vector
RF / SVM / XGBoost
Representation-learning approach
Learned representations
FM embeddings (Fitbit)
LLM text (EMA, context)
Concat / gated /
cross-attention
one fused vector
Linear head
(frozen encoders)
Binary risk
PSS ≥ 14
ZBI ≥ 21
UCLALS ≥ 35
high vs. low
Evaluation: participant-level LOOCV · AUC, accuracy, F1, recall · baselines: majority class, random vote
Supplementary: source ablation · phone-sensing subset · CARE longitudinal analysis